

[illegible]

***"Strength is Life, Weakness is Death."* – Swami Vivekananda**

IoT Framework



- Framework provides a **development environment**.
 - It provides appropriate infrastructure & tools to design and implement the architecture
- IoT framework comprises of **large number of components**
 - sensors, sensor systems, gateways, mobile app, embedded controller, data management platform, analytical platform, and so on.
 - support **interoperability** among all devices, provides **secure connectivity**, **reliability** in data transfer, **interface** to 3rd party application to built on it, and **many more**.

Few IoT Framework	Few IoT Framework
RTI (Real-Time Innovations) Connex DD	Cisco Ultra IoT
Salesforce IoT cloud	Azure IoT
Eclipse IoT	ThingWorx
GE (General Electronic) Predix	AWS IoT
Watson IoT	Kaa

Cisco Ultra IoT

- ✓ It is a fully virtualized, 5G-ready services core
- ✓ It supports the 3GPP Cellular IoT (CIoT) architecture
- ✓ It also supports wide range of IoT solutions across vertical markets
- ✓ It includes core network support for all wireless IoT connectivity (e.g. 2G/3G/LTE, NB-IoT & CAT-M1, WiFi, LoRaWAN, etc.)

- ❑ Cisco relies on three major components:
 - ✓ Virtualized data centre
 - ✓ Intelligent network
 - ✓ Connected devices

- ❑ Cisco Ultra IoT is a part of Cisco Ultra Service Platform



Source: <https://www.cisco.com/c/en/us/solutions/collateral/service-provider/ultra-services-platform/cisco-ultra-iot-at-a-glance.html>

Cont...

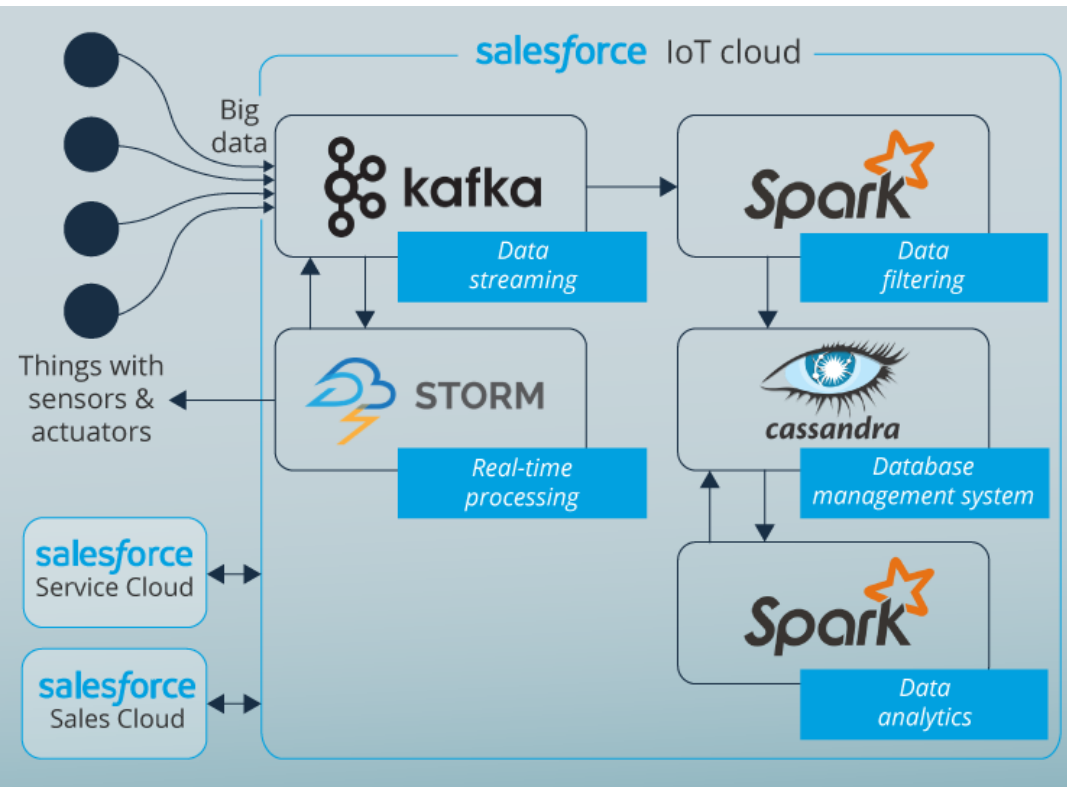


- It delivers **fully-virtualized mobile core functions** that bring **benefits to Service Providers**:
 - **Simplifies and automates services provisioning** with Cisco's industry-leading orchestration solutions
 - Provide **end-to-end 5G network slicing**, which uses powerful provisioning and management capabilities to enable each enterprise customer or IoT Service **to get its own secure, highly reliable, dedicated virtualized mobile core**
 - **Separates the user plane from the control plane** with a distributed architecture using SDN, allowing traffic to go directly to the Internet without costly backhaul
 - ✓ **Cisco CIoT Serving Gateway Node (CSGN)**: it terminates multiple access technologies
 - ✓ **Cisco enhanced Service Capability Exposure Function (eSCEF)**: associates endpoints with applications and exposes network capabilities and data to applications through an open API.

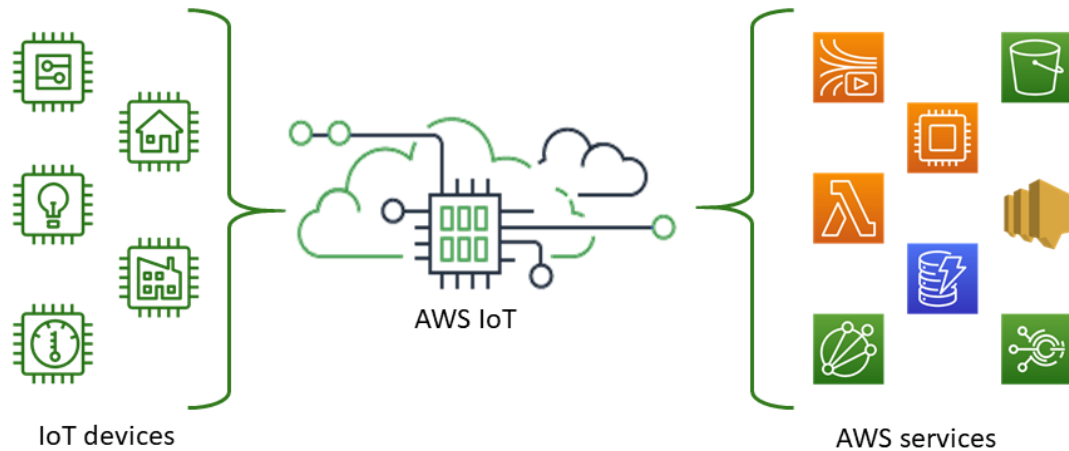
□ **Benefits/Goal at-a-glance:**

- **Grow revenues** by securely connecting IoT devices and sensors over a multi-access core with unified policy, charging, and service capabilities
- **Monetize network intelligence and services** by enhancing capabilities of IoT applications
- Meet the scaling requirements of **Massive IoT** with a fully virtualized 5G-ready mobile core solution

Salesforce IoT Cloud



- Salesforce IoT Framework
 - Combines **IoT** and **big data**
- A new data is sent to **Kafka**, where **Spark** is responsible to transmit it to **Cassandra** for further processing.
- Meanwhile, **Storm** analyses the dataset to formulate responses for current events
- The **backbone** of these technologies:
 - Salesforce's Heroku PaaS
 - AWS Cloud
 - Apache Technologies and Thunder event-processing engine



- AWS IoT provides **cloud services**
 - that connect the IoT devices to other devices and AWS cloud services.
- AWS IoT provides **device software**
 - that facilitate development and integration of the IoT devices with AWS IoT-based solutions

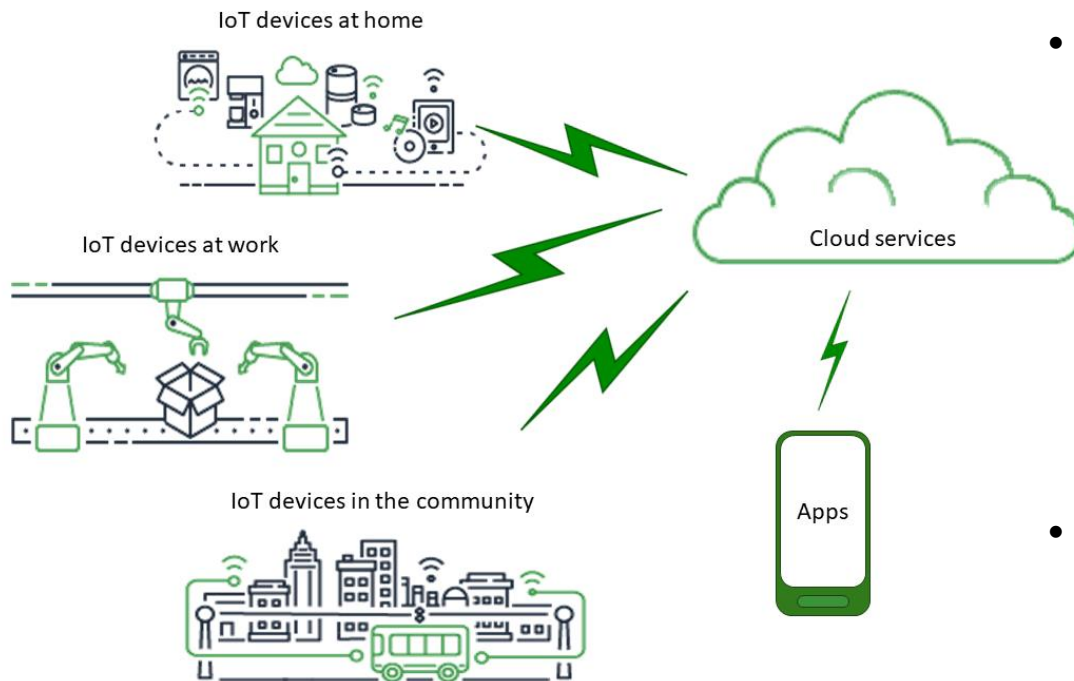
AWS IoT Interfaces



How the IoT end devices and apps access AWS IoT:

- **AWS IoT Device SDKs**
 - Build applications on your devices that send messages to and receive messages from AWS IoT.
- **AWS Command Line Interface (AWS CLI)**
 - Run commands for AWS IoT on Windows, MacOS, and Linux.
 - These commands allow you to create and manage thing objects, certificates, rules, jobs, and policies.
- **AWS IoT API**
 - Build your IoT applications using HTTP or HTTPS requests.
 - These API actions allow you to programmatically create and manage thing objects, certificates, rules, and policies.
- **AWS SDKs**
 - Build your IoT applications using language-specific APIs.
 - These SDKs wrap the HTTP/HTTPS API and allow you to program in any of the supported languages.

How AWS IoT works



- **Cloud services**

- Cloud services are distributed, large-scale data storage and processing services that are connected to the Internet.
- **e.g.** IoT connection and management services (*AWS IoT*)
- **e.g.** Compute services (*AWS Lambda*)
- **e.g.** Database services (*Amazon DynamoDB*)

- **Communications**

- Devices communicate with cloud services by using various technologies and protocols.
- **e.g.** Wi-Fi/ Broadband Internet, Broadband/ Narrowband cellular data

- **Interfaces**

- An interface is a component that connects a device to the physical world.
- **e.g.** User interfaces, Sensors, Actuators

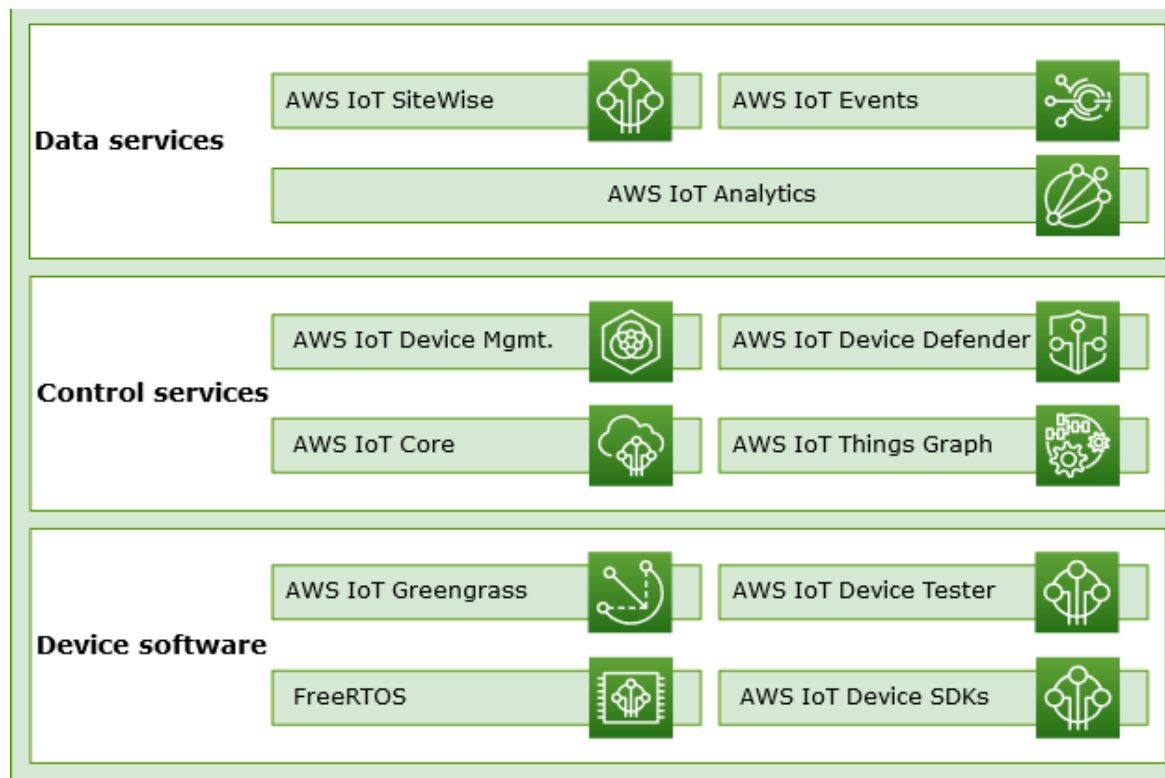
- **Apps**

- Apps give end users access to IoT devices and the features provided by the cloud services to which those devices are connected.

- **Devices**

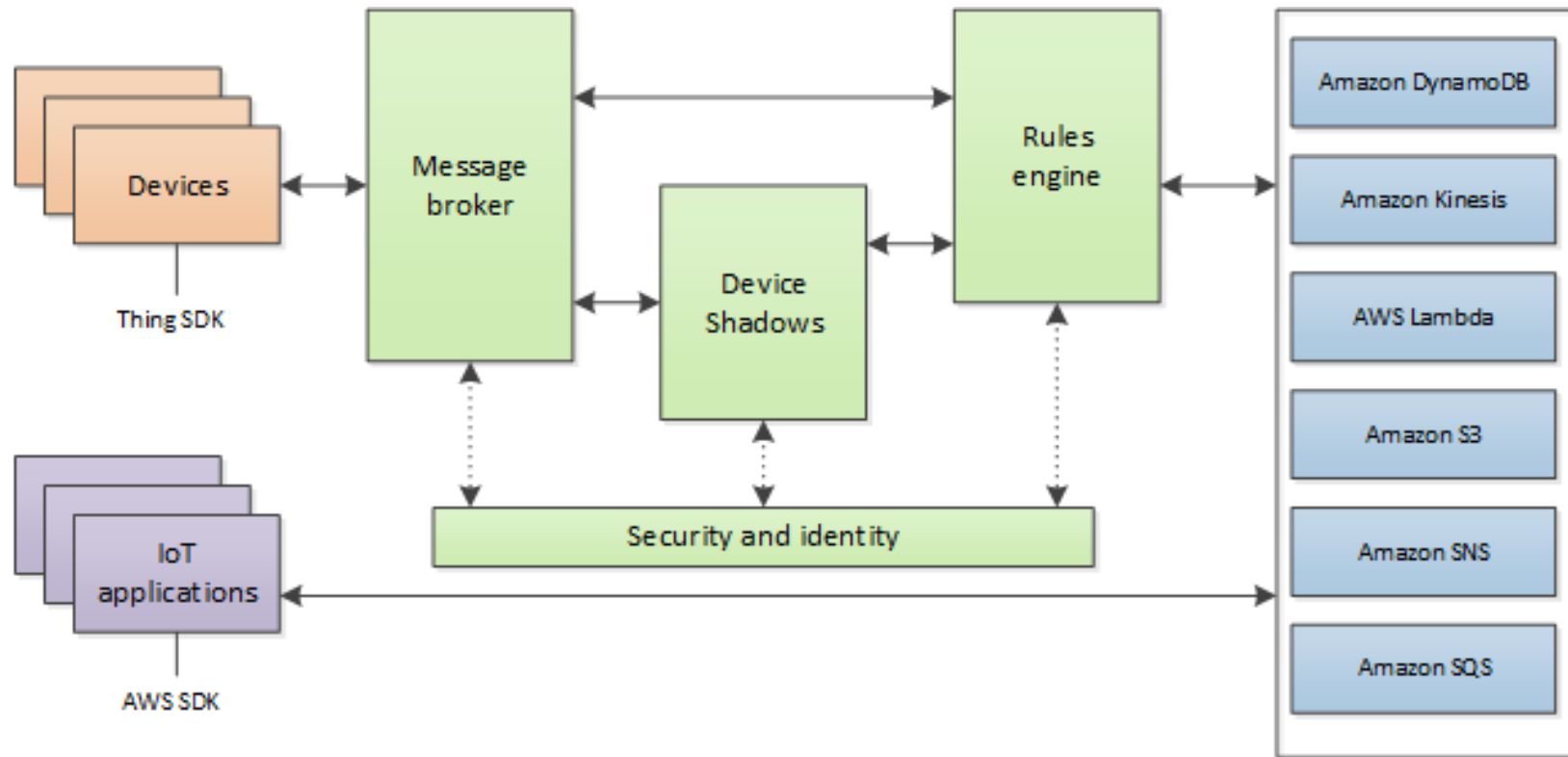
- A device is a type of hardware that manages interfaces and communications.
- **e.g.** Raspberry Pi, Arduino, Custom IoT devices

AWS IoT services



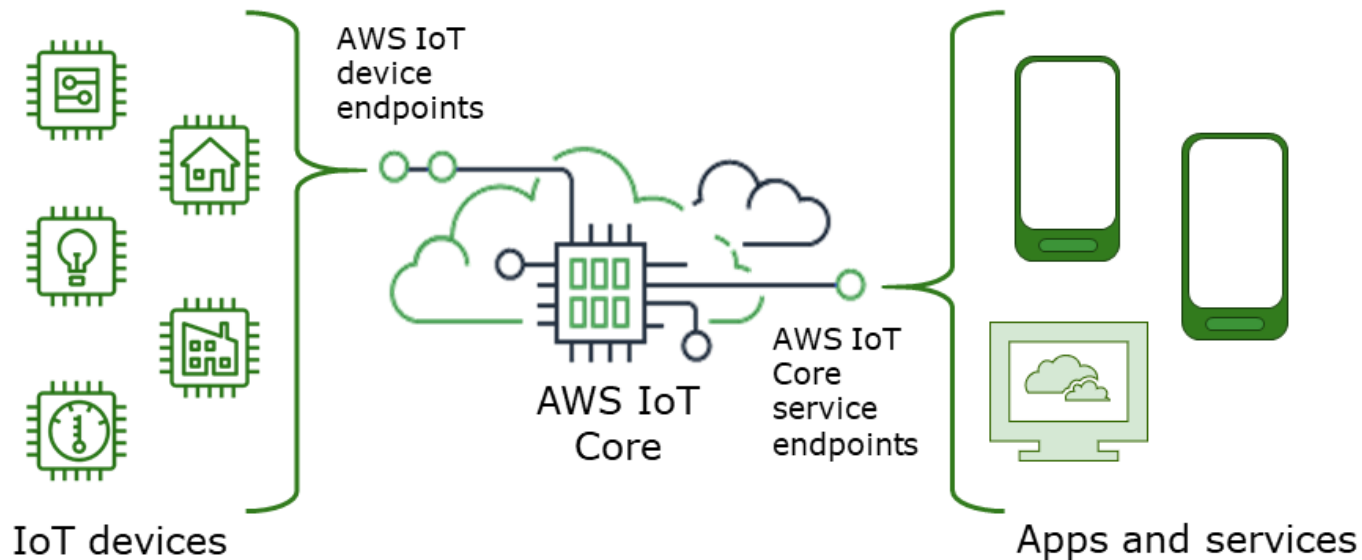
- **AWS IoT Greengrass** extends AWS to edge devices so they can act locally on the data they generate and use the cloud for management, analytics, and durable storage.
- **AWS IoT Core** is a managed cloud service that enables connected devices to securely interact with cloud applications and other devices.
- **AWS IoT Analytics** filters, transforms, and enriches IoT data before storing it in a time-series data store for analysis.

AWS IoT Core services



- AWS [IoT Core](#) provides the services that connect your IoT devices to the AWS Cloud so that other cloud services and applications can interact with your internet-connected devices.
- The [Message broker](#) handles communication between your devices and AWS IoT.
- The [Device Shadow](#) service maintains a device's state so that applications can communicate with a device whether the device is online or not.
- The [Rules engine](#) connects data from the message broker to other AWS services for storage and additional processing.

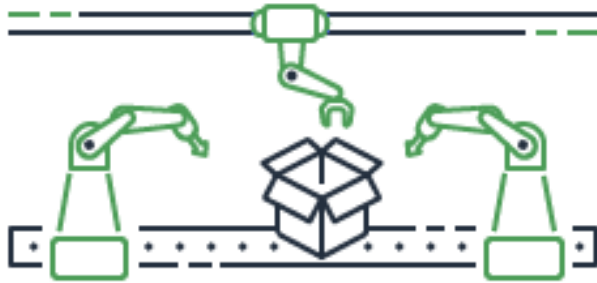
Connecting to AWS IoT Core



- AWS **IoT Core** supports connections with IoT devices, services, and apps.
- **Devices** connect to the AWS IoT Core so they can send data to and receive data from AWS IoT services and other devices.
- **Apps** and other services also connect to AWS IoT Core to control and manage the IoT devices and process the data from your IoT solution.
- In general, the two ways to interact with AWS IoT are through the **AWS IoT Core service endpoints** and the **AWS IoT device endpoints**.

What AWS IoT can do? - Example

IoT in Industry



- Use AWS IoT to build predictive quality models in industrial operations
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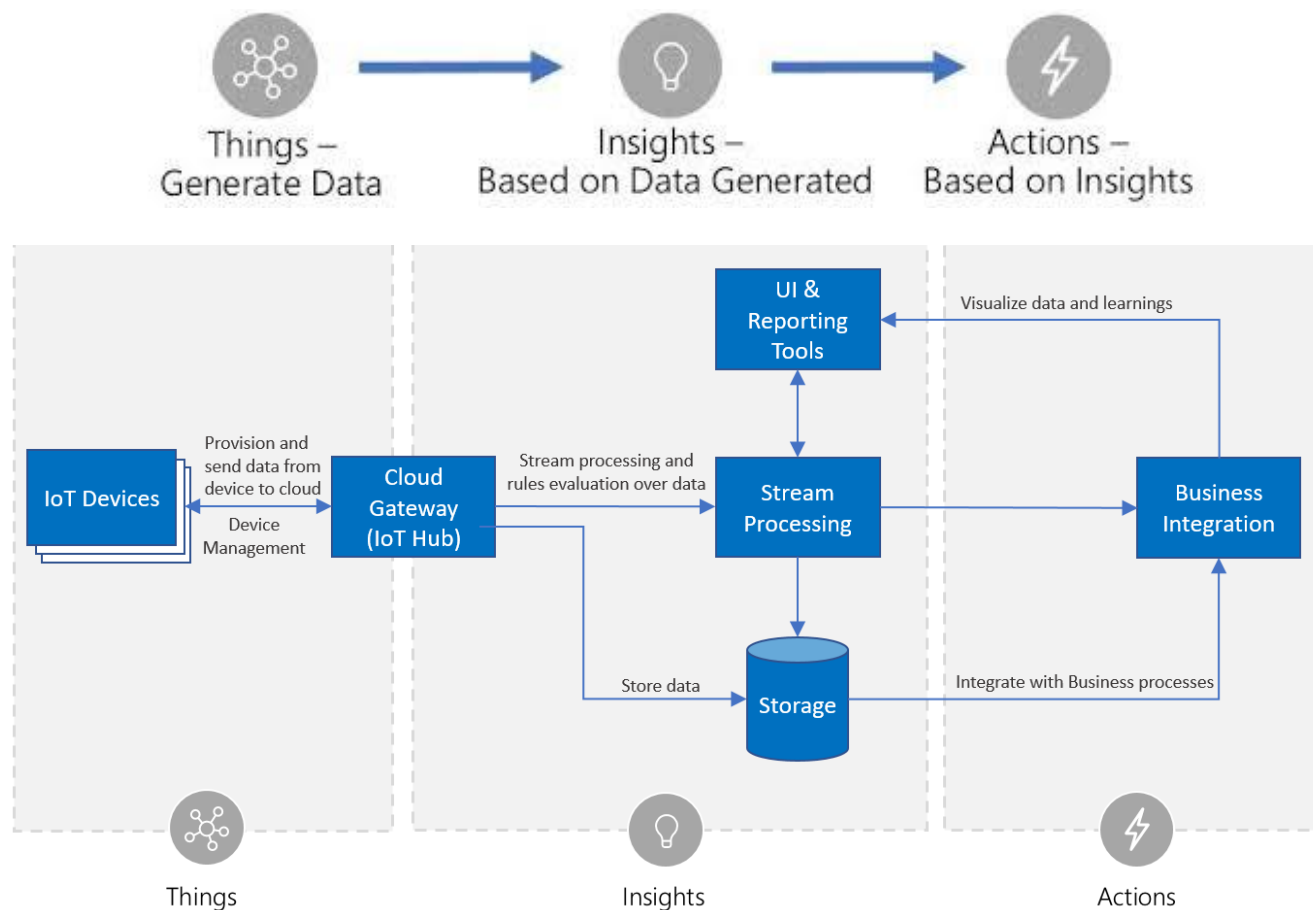
IoT in Home automation



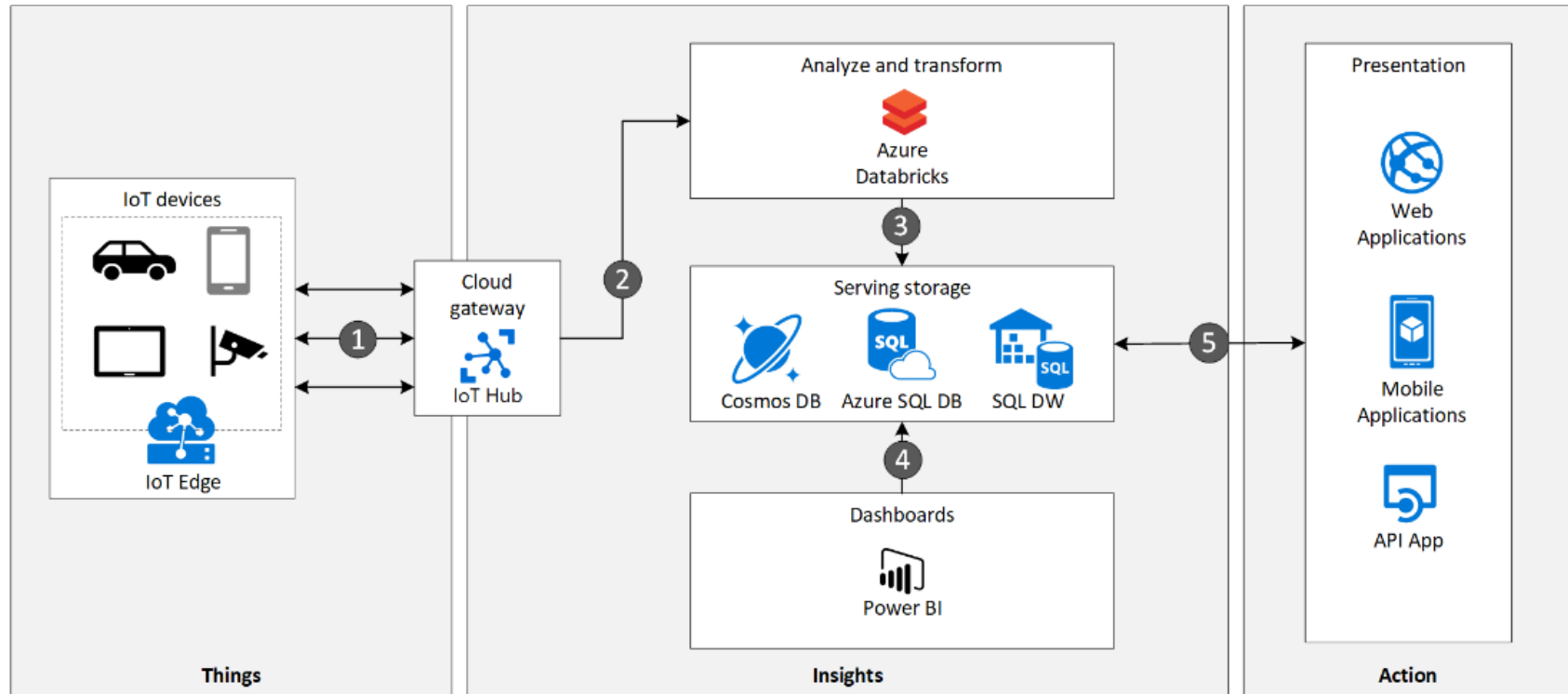
- Use AWS IoT in your connected home
- Use AWS IoT to provide home security and monitoring

Azure IoT

- Azure IoT is a **collection of Microsoft-managed cloud services** that connect, monitor, and control billions of IoT assets.
 - It **deals with how these services work together across three components** – Things, Insights, Actions



Azure IoT Sample Architecture



- **Things:** The **physical objects or things**, such as industrial equipment, devices or sensors, that connect to the cloud persistently or intermittently.
- **Insights:** Information collected by the things, which is **analysed and turned into actionable knowledge** either by people or AI.
- **Action:** The way **people respond to those insights** and connect them to their business, as well as the systems and tools they use.

Cont...



- **Things:**

- Secure your edge devices and hardware
 - Azure Sphere and RTOS
 - [Windows IoT](#)
- Simplify connectivity with certified devices
 - IoT Plug and Play
 - Azure Certified Devices
- Push applications and workloads to the edge
 - Azure IoT [Edge](#)
 - Azure Stack
 - Azure Stack Edge

- **Insights:**

- Connect and manage your device data with flexibility
 - Azure IoT [Hub](#)
 - Azure IoT [Central](#)
- Transform your data for interoperability
 - Azure APIs
- Create digital models of entire environments
 - Azure Digital Twins

- **Action:**

- Integrate with your business applications
 - Power BI
 - Dynamics 365
 - CRM
 - Field Services
- Security from edge to cloud
 - Azure [Defender](#) for IoT
- Build connected ecosystems

- **Azure IoT Hub and Digital Twins** provide the building blocks for companies
 - to construct customised solutions for complex IoT scenarios.
- Enable **highly secure and reliable communication** between your IoT application and the devices it manages.
 - Security-enhanced communication channel for sending and receiving data from IoT devices
- Provides a **cloud-hosted solution back end** to connect virtually any device.
- **Extend** your solution from the cloud **to the edge** with per-device **authentication**
- **Built-in device management and provisioning** to connect and manage IoT devices at scale
- Compatibility with Azure IoT Edge and Azure Stack for **building hybrid IoT applications**
- **Azure Digital Twins:**
 - Digital representation of your connected device.
 - It's a JSON file in the cloud managed by IoT Hub, and there is one of them for each of your IoT devices connected to IoT Hub.

Example: Azure IoT for healthcare

Enabling the next generation of healthcare IoT solutions

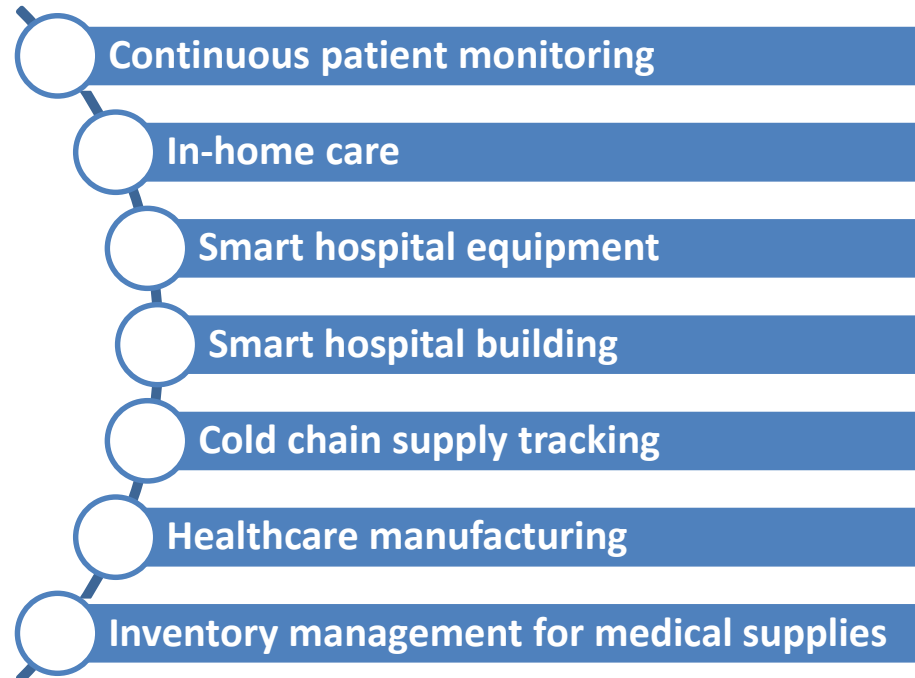
- Deliver personalised care,
- Empower care teams and employees, and
- Improve operational outcomes.

Seamless

Smart

Secure

Applications



Thanks!



Figures and slide materials are taken from the following:

1. Mayur Ramgir, “**Internet of Things: Architecture, Implementation and Security**”, 1st Edition, 2020, Pearson India.
2. <https://docs.aws.amazon.com/iot/latest/developerguide/what-is-aws-iot.html>
3. <https://docs.microsoft.com/en-us/azure/iot-fundamentals/>